রাঙ্গামাটি বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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Lab Report On: System Analysis and Design of RMSTU Canteen Management System

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ABSTRACT

The purpose of this system analysis is to develop a proper Canteen Management System of RMSTU .

Nowadays people do not have much time to spend in canteen by just waiting for the waiter to take their order. Many customers visit the canteen in their lunch break and recess, so they have limited time to eat and return to their respective offices and university classes. So, this system will help them to save time and order food whenever they want without calling the waiter again and again.

Manual system involves paperwork in the form of maintaining various files and manuals. Maintaining critical information in the files and manuals is full of risks and tedious process. Including a framework showing how to apply internet technology progressively as skills and confidence grow, the project demonstrates the routes from adapting materials to developing an online environment.

This canteen automation system enables the end users to register online, read and select the food from the e-menu card and order food online by just selecting the food that the user want to have. This system will assist managers and administrators in managing canteen effectively by reducing the workload in the present system and reducing time delay in data processing. Moreover, the new system will focus on maximizing the profit and allows the managers to track sales, inventory levels, and staff performance. The Canteen Management system aims to improve canteen facilities, increases sells and profits, enhance customer satisfaction, and reduce operational costs.

We all worked together in this project. All of the members of ‘Scienceholic’ contributed same efforts, energy, ideas and thoughts in each process of this work. Everyone was presented during the whole teamwork.

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Problem Recognition:

- **Morning Snacks**: Lack of availability of light snacks in the morning.
- **Limited Food Variation**: Limited variation in food options.
- **Dietary Restrictions and Health Concerns**: Teachers and officials with health-related dietary restrictions face challenges in finding suitable food options.
- **Pricing and Food Quality**: High food prices with unsatisfactory food quality.
- **Inadequate Space and Capacity**: Small cafeteria with limited seating capacity.
- **Hygiene and Sanitation Concerns**: Absence of proper handwashing facilities , absence of washroom facilities in the cafeteria , and hygiene issues.
- **Lack of Skill and Low Service Quality**: Workers lack necessary skills, resulting in low service quality.
- **Ordering Convenience**: Customers prefer ordering from their seats due to tiredness or convenience.
- **Availability and Timing:** Students find it challenging to secure their preferred food items during lunchtime due to limited availability.
- **Service Delays and Waiting**: Customers experience delays in placing order and service.
- **Quantity vs. Price:** Inadequate portion sizes compared to the price.
- **Cafeteria Location:** The cafeteria's location is distant from the offices of university officials.
- **Presentation and Guest Hosting**: Insufficient facilities for hosting guests or conducting presentations.
- **Funding and Profitability Challenges:** The cafeteria does not receive financial support from the university.
- **Vacation Period Sales**: Reduced sales during vacations impact revenue.

- **Evening Sales**: Decline in sales of food items after evening hours.
- **Rent Expenses**: Cafeteria shareholder's have to pay rent, impacting overall finances.

Feasibility Study:

Technical Feasibility:

- **Online ordering system:** Introducing an online ordering system such as a proper websites or a software application , offers convenience, reduced wait times, and customization for users. Orders can be placed remotely, cutting queue times and enabling personalized menu choices. Real-time tracking, integrated payments, and order updates enhance user satisfaction. This solution aligns with our goal to enhance accessibility and satisfaction.
- **Skilled Staff Recruitment:** By recruiting staff members who possess the necessary technical competencies, a smooth integration of the proposed enhancements and the ongoing efficient operation of our cafeteria system can be ensured. As they can contribute to the successful implementation, seamless operation, and maintenance of the new components, ensuring a smooth user experience and optimized cafeteria operations.
- **Payment system:**Cafeteria management team should introduce payment system using mobile banking such as bkash, rocket, nagad so that customers can pay for their food even if they are out of cash.

Operational Feasibility:

- **Recruiting staff:** Cafeteria management team should consider the recruitment of additional staff to enhance cafeteria operations. Hiring more staff impacts workload distribution, user service, and

overall efficiency. We assess the feasibility of this approach and its alignment with existing operational processes and resources.

- **Expanding Food Variety :** Introducing new food items impacts sourcing, kitchen operations, and responding to user preferences, leading to higher sales. We assess the feasibility of this initiative, ensuring it aligns harmoniously with our existing operations, available resources, and potential customer demand.
- **Staff Training:** Staff training is crucial to ensuring that the staff can effectively adapt to new processes, technologies, or changes in operations. It enables the team to operate efficiently, provide optimal user support, and provide better service to the customer.
- **Comfortable environment and increased seating** : A comfortable environment and increased seating capacity positively influences user satisfaction. By expanding seating, cafeteria can accommodate more users and promote a relaxed dining experience.
- **Discount Scheme**: Introducing occasional discounts can attract customers, boost sales, and improve the cafeteria's value proposition. We assess the feasibility of this approach by evaluating its alignment with pricing strategies, user preferences, and financial sustainability.

Economic feasibility:

- **Raw Food Cost**: The cost of raw food materials directly affects the cafeteria's financial viability. Assessing the feasibility of sourcing ingredients locally or exploring potential cost-saving measures can lead to optimized procurement strategies. By understanding the correlation between raw food expenses and the cafeteria's budget, we ensure efficient allocation of resources and sustainable financial outcomes.
- **Transportation Cost:** Due to the distance between the university cafeteria and the local markets of Rangamati Sadar, the transportation cost for food is considerably high. Consequently,

there's a need to explore options with nearby food vendors or suppliers in the vicinity of the university.

- **Price structure:** Recognizing the concern that cafeteria food prices may not align with the quantity offered, we aim to strike a balance that satisfies customers and encourages increased patronage. By evaluating potential pricing changes, we seek to improve the cafeteria experience, ensuring that customers find the value they seek without compromising the financial sustainability of the cafeteria.

- **Electricity and gas bills**: The cafeteria is grappling with rising electricity and gas expenses, which are becoming increasingly challenging to manage. The growing bills of electricity and food processing are putting a strain on resources. However, it's important to note that this issue is beyond our individual control; it's a nationwide concern rather than a personal one.

System Analysis:

Percentage of Customers:

- 55% Customers are students.
- 25% Customers are teachers.
- 18% Customers are officials , workers ,and stuffs.
- 2% Customers are tourist .

Raw Food and other Food ingredient:

1. Vegetables, fishes, meats others from local market.
2. All spices from town.

Existing System:

Total staff number: 5

- Manager: 1
- Kitchen staff: 1
- Waiter staff: 3

Serving capacity:

- Sitting space for teachers and officials: 12 customers at a time.
- Sitting space for students: No specific sitting space.

Cafeteria space: 400 square feet approx.
Space cost: Rental cost (8000/-)
Share business: (6 person)

Technological Support in Promotion:

1. No business website is available .
2. Facebook page is not available.

Staff Details:

Department	Current Salary	No of Staff
Kitchen	22,000/-	1
Waiter	10,000/- 4000/- 3000/-	3
Manager	26,000/-	1

Monthly Staff Salary:

- 1 Professional Cooks = 22,000/-
- 1 Cashier = 10,000/-
- 2 Waiter = 7,000/- (3,000+4,000)
- 1 Manager = 26,000/-

- Total Salary Cost = 65,000 /-
- Rent = 3,000/-
- Electricity Bill = 3,000 /-
- Tax = 0 /-

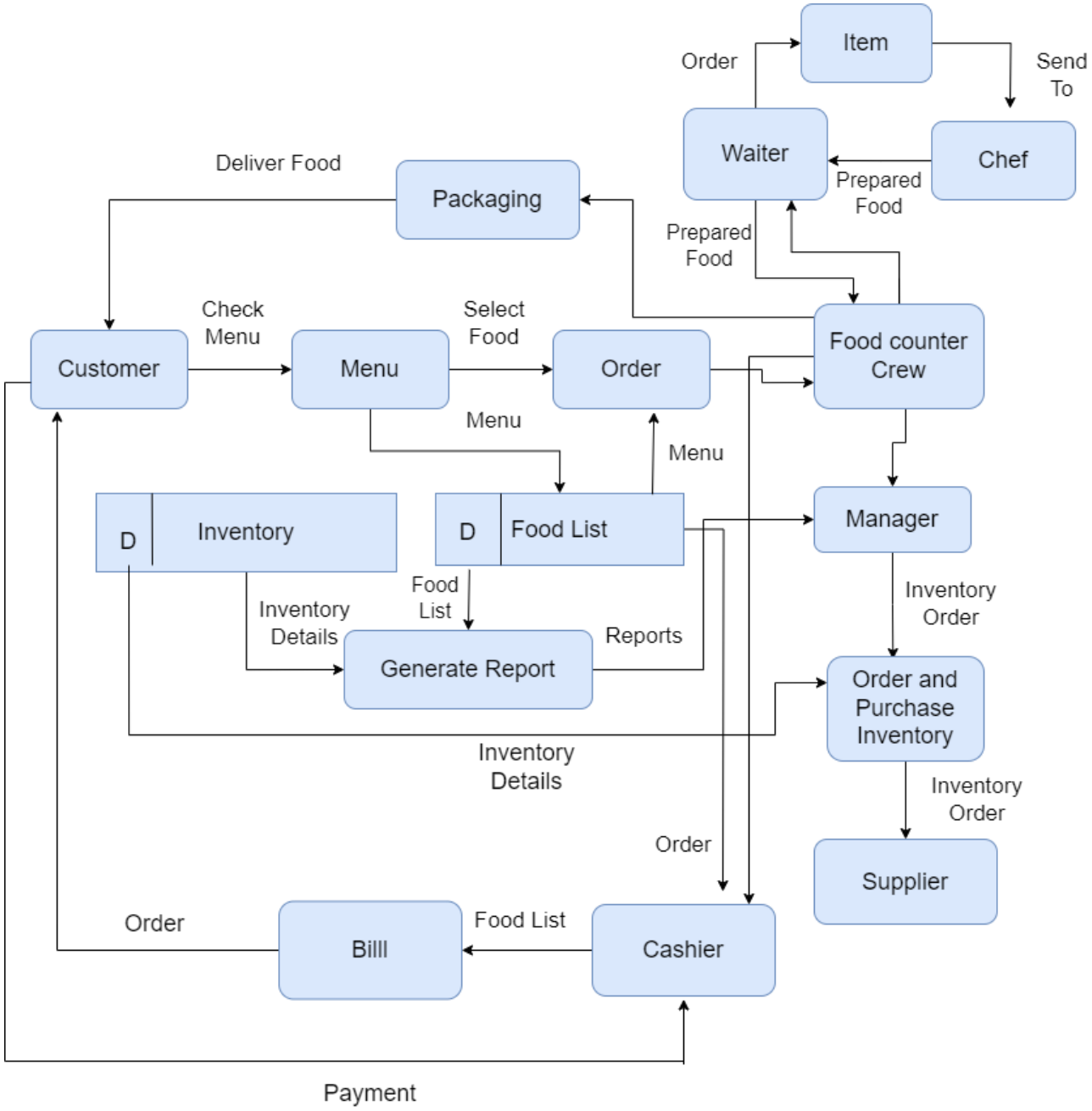
Total Cost = 71,000/-

Weekly Sales Report:

Category	Monday	Tuesday	Wednesday	Thursday
Snacks Food	2900/-	3000/-	2500/-	2300/-
Meal	4,200/-	3,500/-	4,500/-	4,000/-
Juice and Drink	1500/	1000/-	1700/-	1800/-
Total Sells	8,600/-	7,500/-	8,700/-	8,100/-

Monthly Worker Salary and other cost : 71,000/-
Monthly Food Cost : 70,000 – 80,000/-(approx.)
Total Monthly Cost : 146,000/- (approx.)
Monthly Sales : 130,000 – 150,000/-
Monthly avg. Profit : 6000/- (approx.)

Block Diagram(Existing System):



System Design :

Designing is the most important and the most efficient function while software development. Without a proper design, it is very difficult to develop an appropriate software that fulfils nearly all user demands. Therefore, managing designing part in an organization is a critical activity. A designer needs to ensure that the design created by him can be easily understood by all the members of developing a team. A proper design will allow the coder to implement the system development planning properly.

Description of Candidate System(Automated Canteen Ordering System):

1. Time saving: The customer can order food and it is also efficient for canteen workers because this system takes lesser time compared to manually based system.
2. Customers will be encouraged to use the hassle-free system, allowing them to place orders from anywhere, even from their workplace or classroom.
3. They can also view the daily menu ,and since the ordering process is entirely automated, there won't be food wastage. Additionally, customers will have the convenience of checking daily menu without visiting the campus canteen, ultimately saving them time and elimination unnecessary hassle.
4. Payment options include a cashless online payment system, as well as the choice to pay with cash.
5. No complication: Major complication for automated canteen ordering system is adding a product or managing products section. In the automated canteen ordering system. No complicated part is involved in managing sections;
6. Cost effective: It is cheaper. You do not have to purchase multiple copies of software to install on multiple computers. Multiple copies often require you to pay multiple licensing fees, but you are not

actually purchasing any software with an online system that is not a concern.

7. Security: Online system are just as secure. Most online automated canteen ordering system programs allow us to create multiple user accounts with various levels of access. Your data is stored on secure, protected servers that feature firewalls and other online security programs.
8. Product perspective : This system will be developed using PHP, HTML, CSS, MySQL, and other web tools.
9. Product functions: The product developed will provide an easy access to the users to manage and alter the automated canteen ordering system to the need.

Use Case Diagram (Automated ordering System):

A use case diagram is used to depict the interaction between the users and the system. It shows the functionalities of the system from a users' point of view and the different activities a user can perform. The below diagram illustrate the activities of the customer and admin , that would be performed by them on new candidate system.

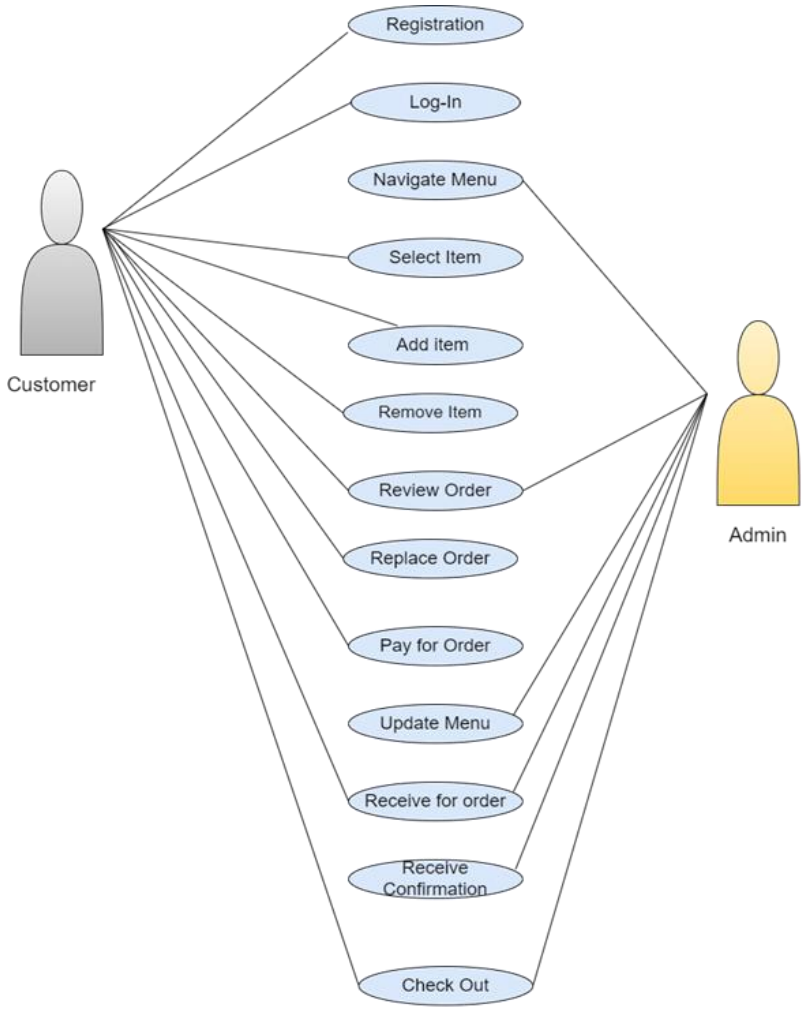


Figure: Use Case Diagram

Control Flow Diagram(Automated Ordering System):

A control flow diagram helps us understand the detail process-it shows us where control starts and ends and where it may branch off in another direction, given certain situations. The below figure illustrate the detail process of the automated ordering system via web-application.

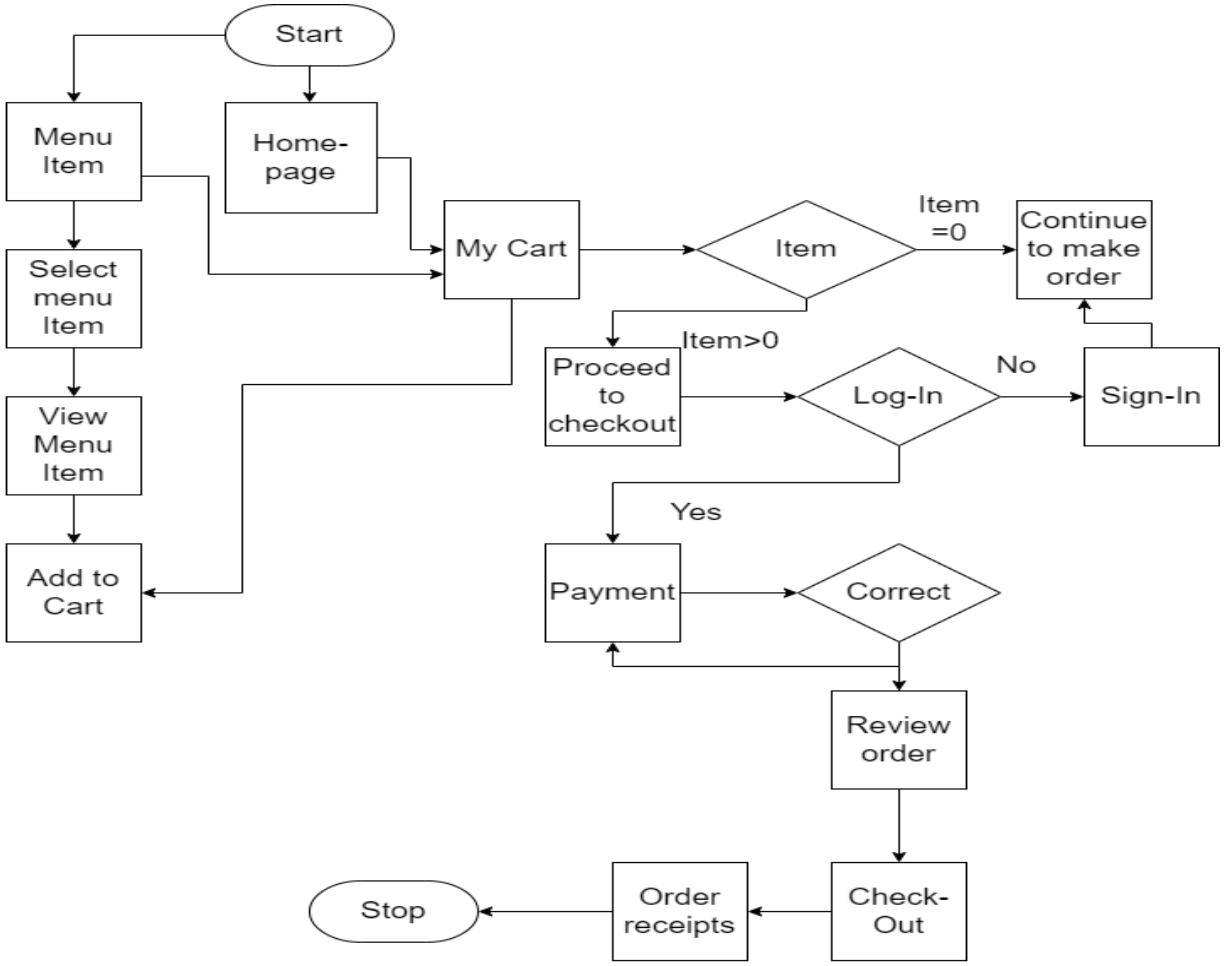
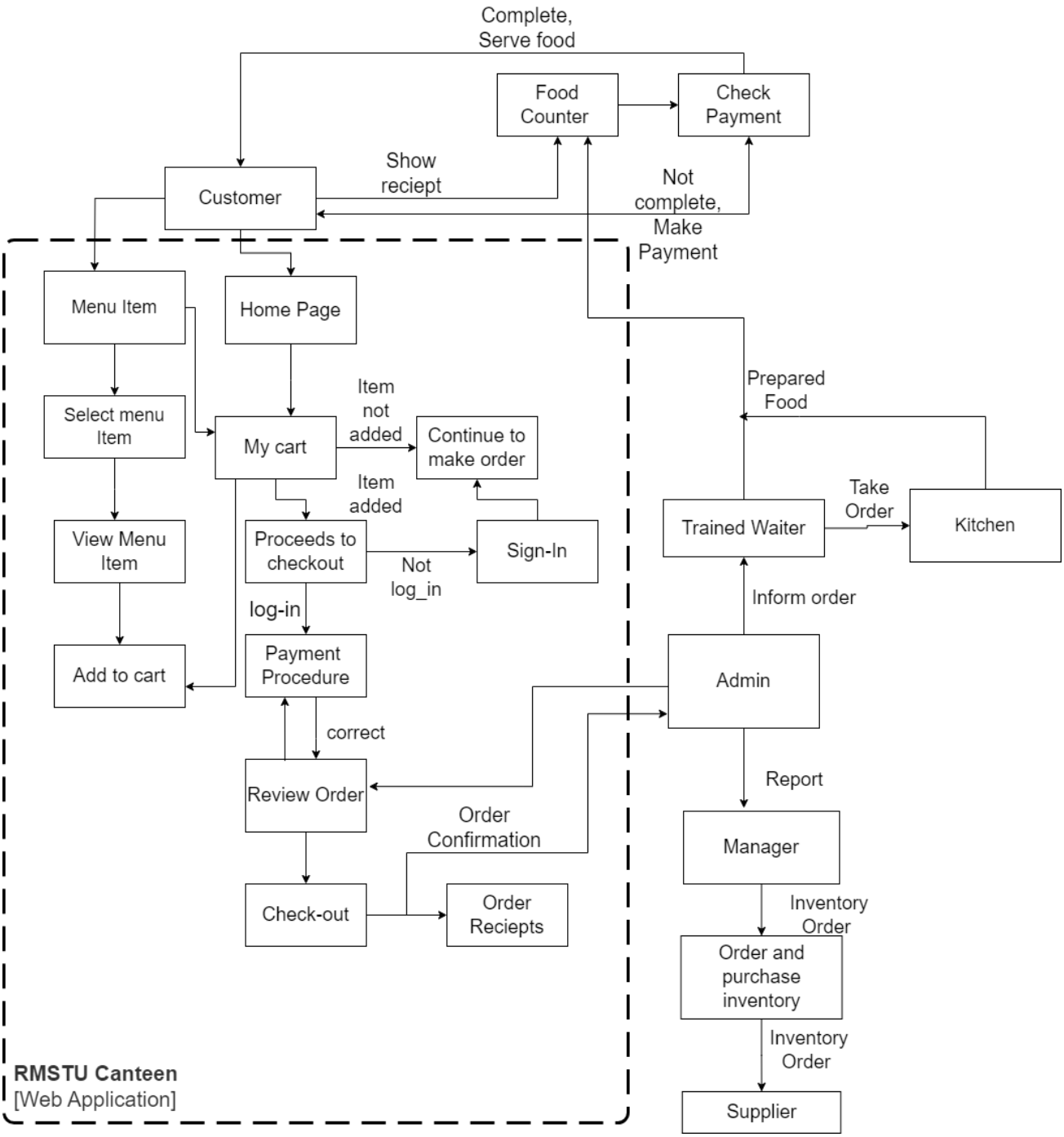


Figure: Control Flow Diagram

Block Diagram (Modifying entire existing System):

Incorporation of the proposed automated ordering system with the existing system:



Sales,Cost & Profit Calculation (Candidate system):

- In this new system, they need to hire an admin to manage the web-application for ensuring a quick –responsive interactive system.

- Their target will be to sell at least 100 platters per day by introducing platter meal option via the new system and regularly updating the menu. As a result, if they make a profit of 15 tk from each order ,then daily 1500 taka and monthly 24000 taka .

- They can ensure customer satisfaction through the online ordering system, as it will alleviate the pressure on their offline service due to constraints in physical pace and a shortage of waitstaff.

- Their sells from new system will be increased if they maintain their service and maintain their food quality. There are no extra costs without the admin's salary.

Sale	Cost
Daily : 100*15 = 1,500/- **Monthly : 1500*16 = 24,000/-**	**Admin's Salary = 18000/-**
Profit : 24000 – 20000 = 4000/- **Which is 66.67% of current profit**	

Comparison Chart:

	Existing System	Proposed system
Profit	Monthly average profit 6000/-	Only from meal monthly profit 4000/- , which is 66.67% of existing profit
Effectiveness	Lack in customer satisfaction, poor ordering system,not effective.	Enhance customer satisfaction, Easy food order processing system, best solution in term of cost ,profit ,effectiveness and sustainability.

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Objective Assessment

Our proposed candidate system is a automated canteen ordering system, which uses a web application that performs fast services to its users. Usually people have to go to canteen and order the foods and they have to wait in queue for a long time to get the orders. But with the help of this users just have to follow a very simple process to order their stuffs.

Limitations and Challenges

This section discusses the problems and challenges faced during the development process. The challenges encountered will be analyzed below.

- Gathering information from canteen staff was challenging.
- Lack of sufficient time.

Future Enhancements

The overall objective of canteen management system is to provide fast service for users of the canteen.
Below are the possible enhancements for canteen ordering system.

- Drive sales with social media: The popularity of mobile phones and the usage of social media platforms using phones is known to the world. So, it is not to be surprise that food chains are now including provisions of using their devices application food plus for ordering to promote their sales.

  • Mobility and ease: With the online payment feature ordering food using a canteen-based application has become a lot easier these days. There occurs no requirement to make use of hard cash. Anyone can order food online using online payment modes right from the canteen ordering app. Customers can also save up payment-related details in their profiles. Therefore, the food ordering script is hassle free, there is no need for adding their account details. With one click, anyone can order food online.

Recommendations

It is known that for any meaningful computer-based information management to be integrated into any organization, proper training and orientation has to be given both to the staff and management. Proper training should be given to the data entry staff on how to handle computer hardware especially sensitive to change in temperature or pressure and as such data can be lost easily. The staff should also be highlighted on the need and advantage of the system and how it will equally assist them in their various field of work. They should also be

informed of the cost of maintaining this new system so that they will handle it with all carefulness. Training materials should be circulated to the personnel. This will at the end generate appreciation and needed interest to operate the system.

Conclusion

We analyzed the current system and found out the problems and based on them proposed a candidate system. We have considered investment, profit and sustainability while designing this system. An automated canteen ordering system is developed where the customers can make an order for the food and avoid the hassles of waiting for the order to be taken by the waiter. Using the application, the end users register online, read the e-menu card, and select the food from the e-menu card to order food online. Once the customer selects the required food item the admin will be able to see the results and start processing the order. This application nullifies the need of a waiter or reduces the workload of the waiter. The advantage is that in a crowded canteen there will be chances that the waiters are overloaded with orders and they are unable to meet the requirement of the customer in a satisfactory manner. Therefore, by using this application, the users can directly place order for food and get it delivered.